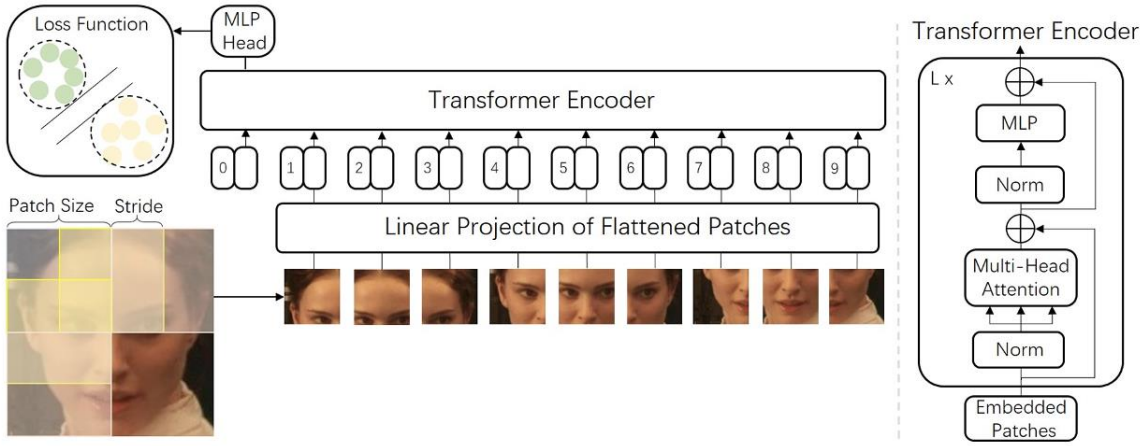
Visión por Computador

Versión 2026



ViT

(CAPÍTULO 3 – CV031)

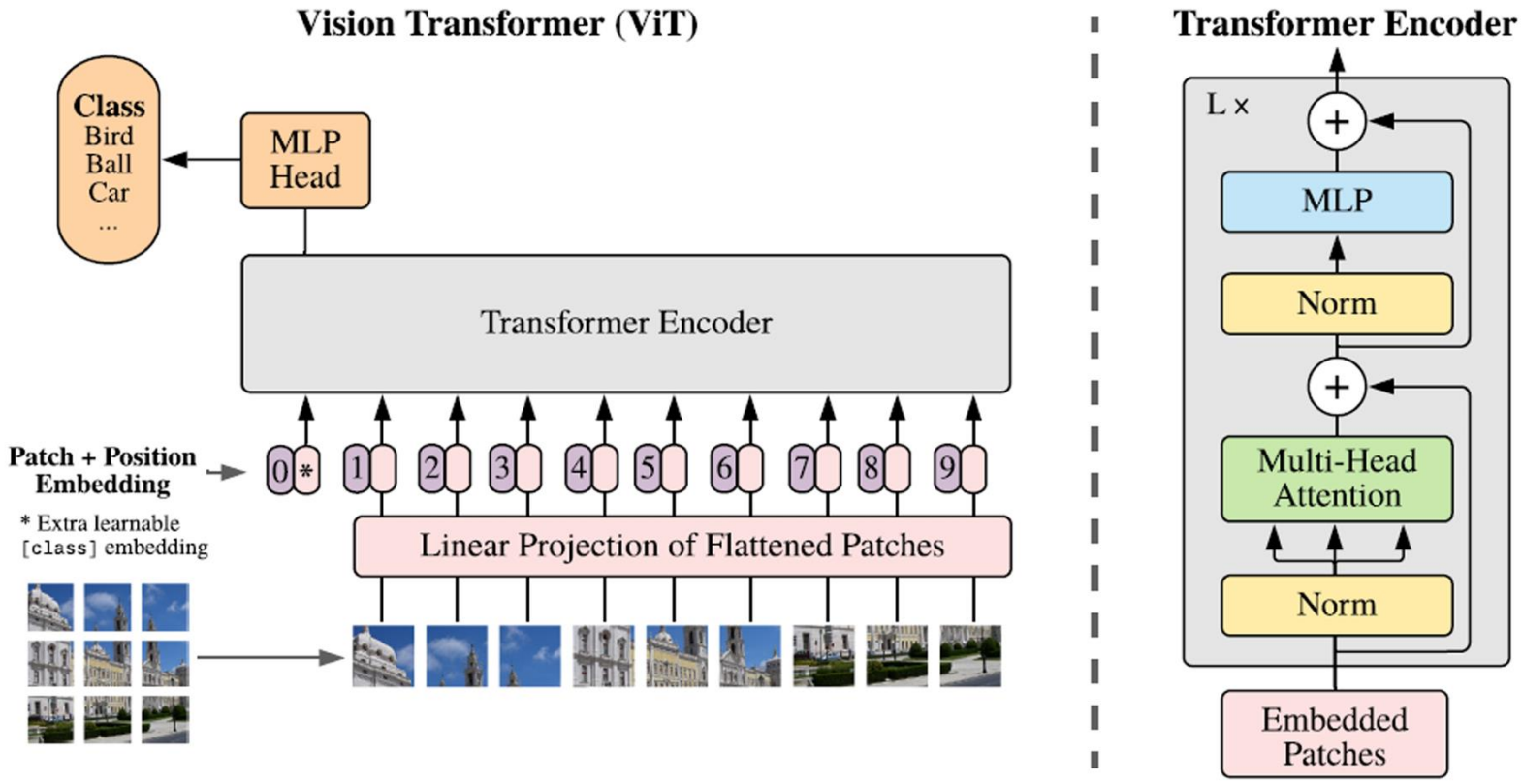


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Visual Transformer (ViT)

Visual Transformer (ViT)



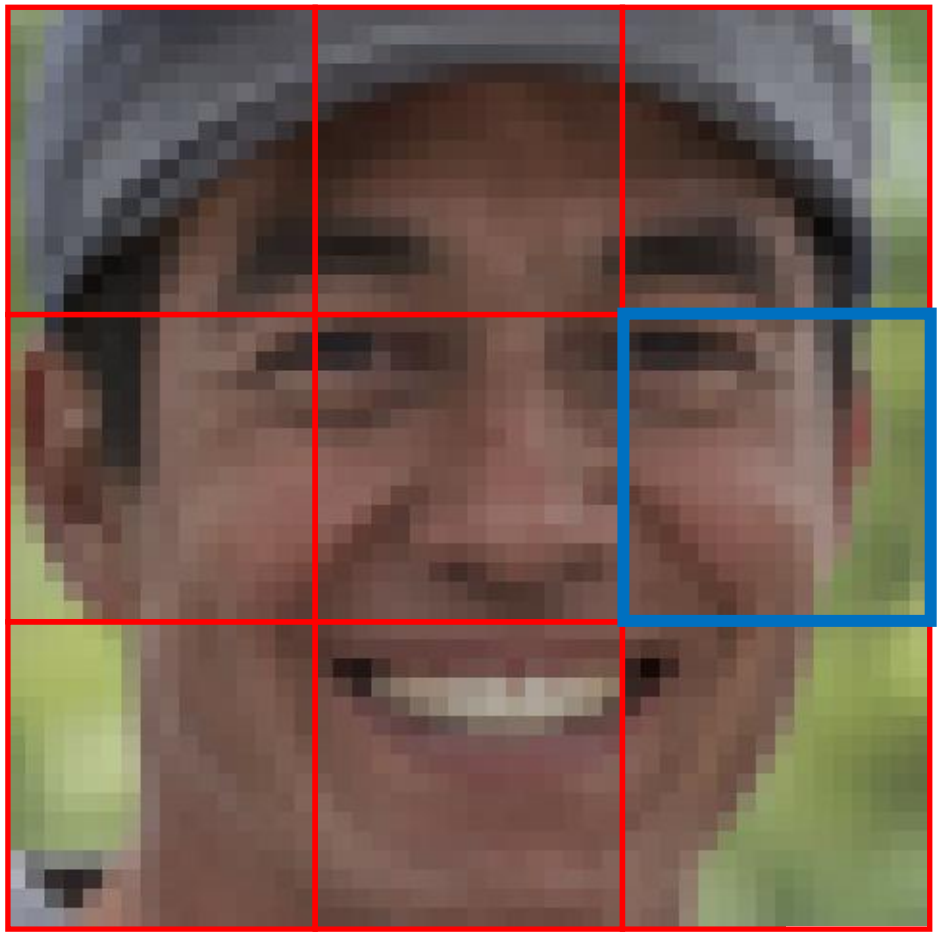
Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani et al. "An image is worth 16x16 words: Transformers for image recognition at scale." arXiv preprint arXiv:2010.11929 (2020).

Visual Transformer (ViT)



Example: ViT in Face Recognition

Visual Transformer (ViT)



48x48

Visual Transformer (ViT)



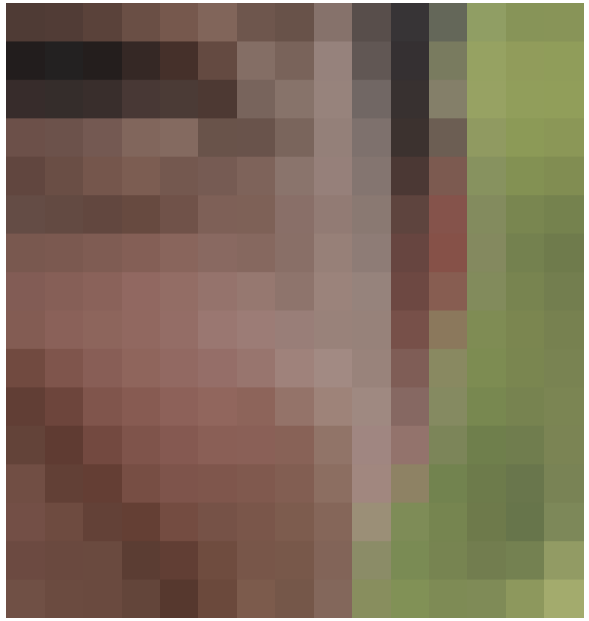
16x16

Visual Transformer (ViT)

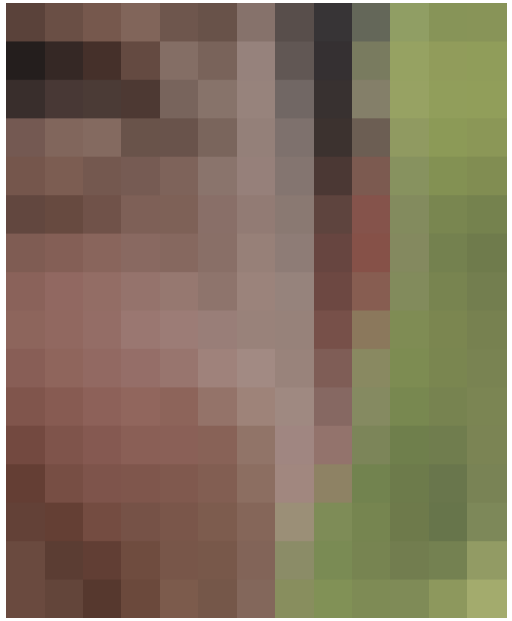


16x16

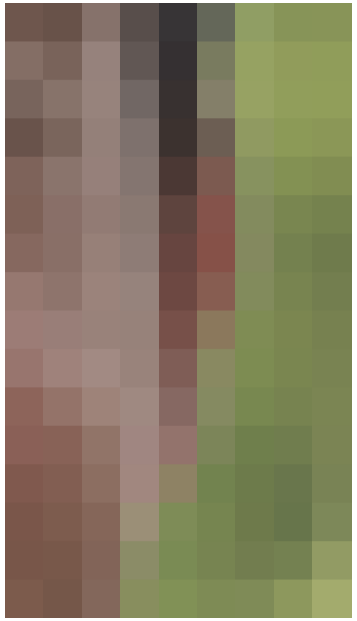
Visual Transformer (ViT)



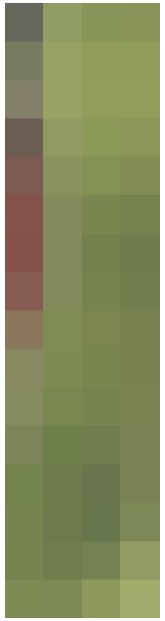
Visual Transformer (ViT)



Visual Transformer (ViT)



Visual Transformer (ViT)



Visual Transformer (ViT)

Visual Transformer (ViT)

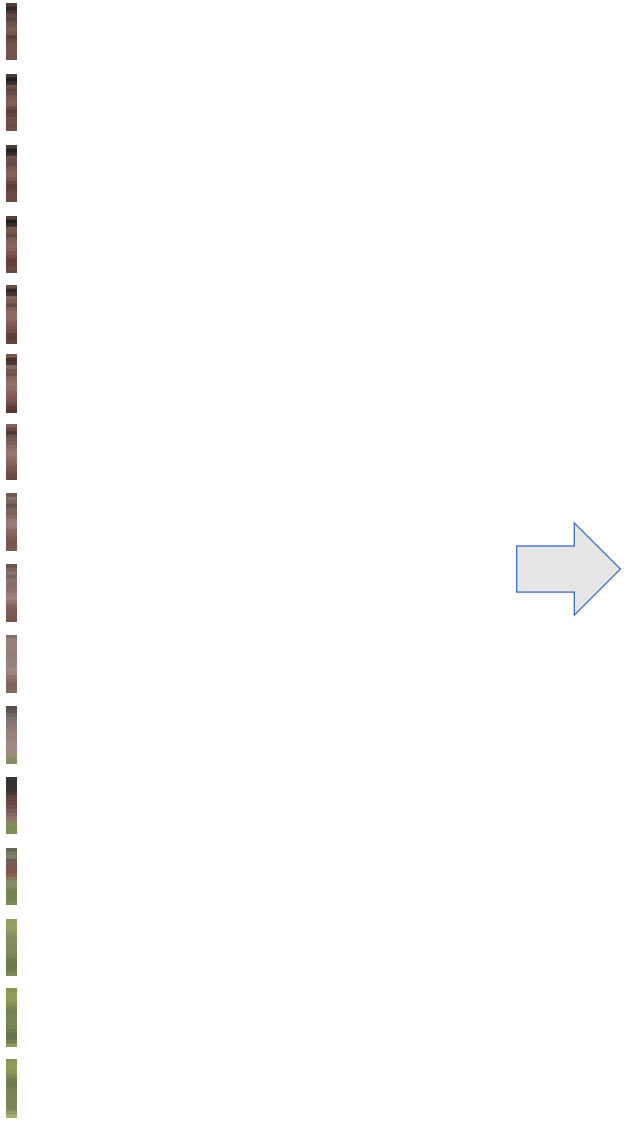
Visual Transformer (ViT)

256x1x3

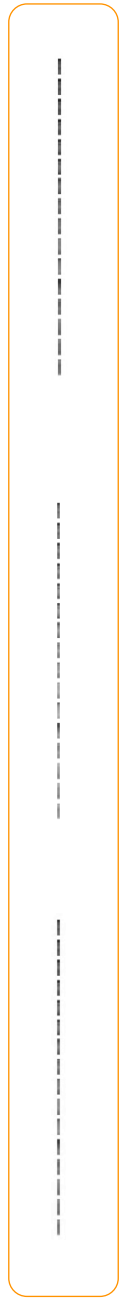
Visual Transformer (ViT)

256x1x3

Visual Transformer (ViT)



256x1x3



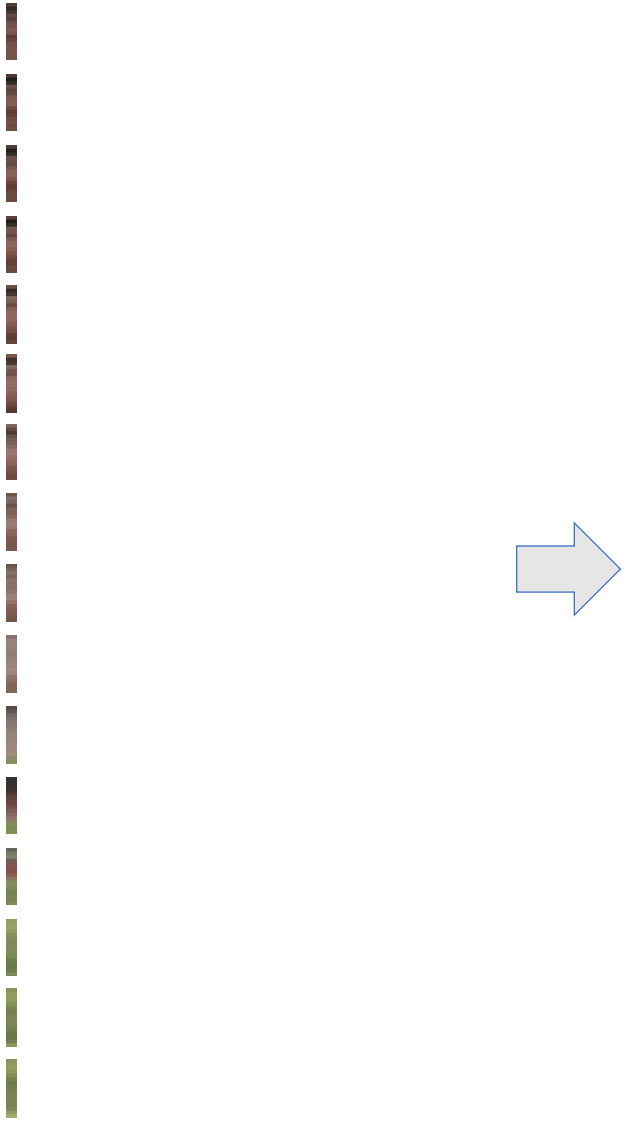
Red
256x1

Green
256x1

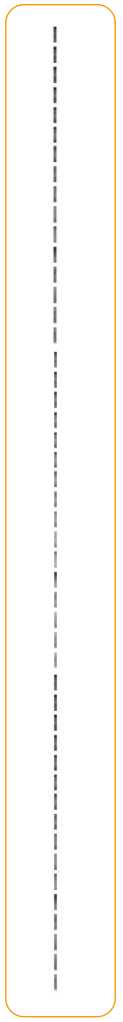
Blue
256x1

768x1

Visual Transformer (ViT)



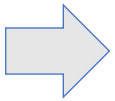
256x1x3

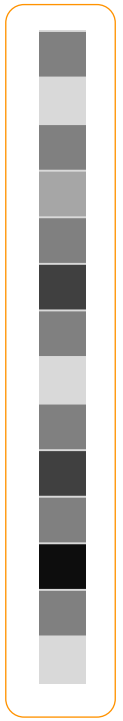


768x1

Visual Transformer (ViT)

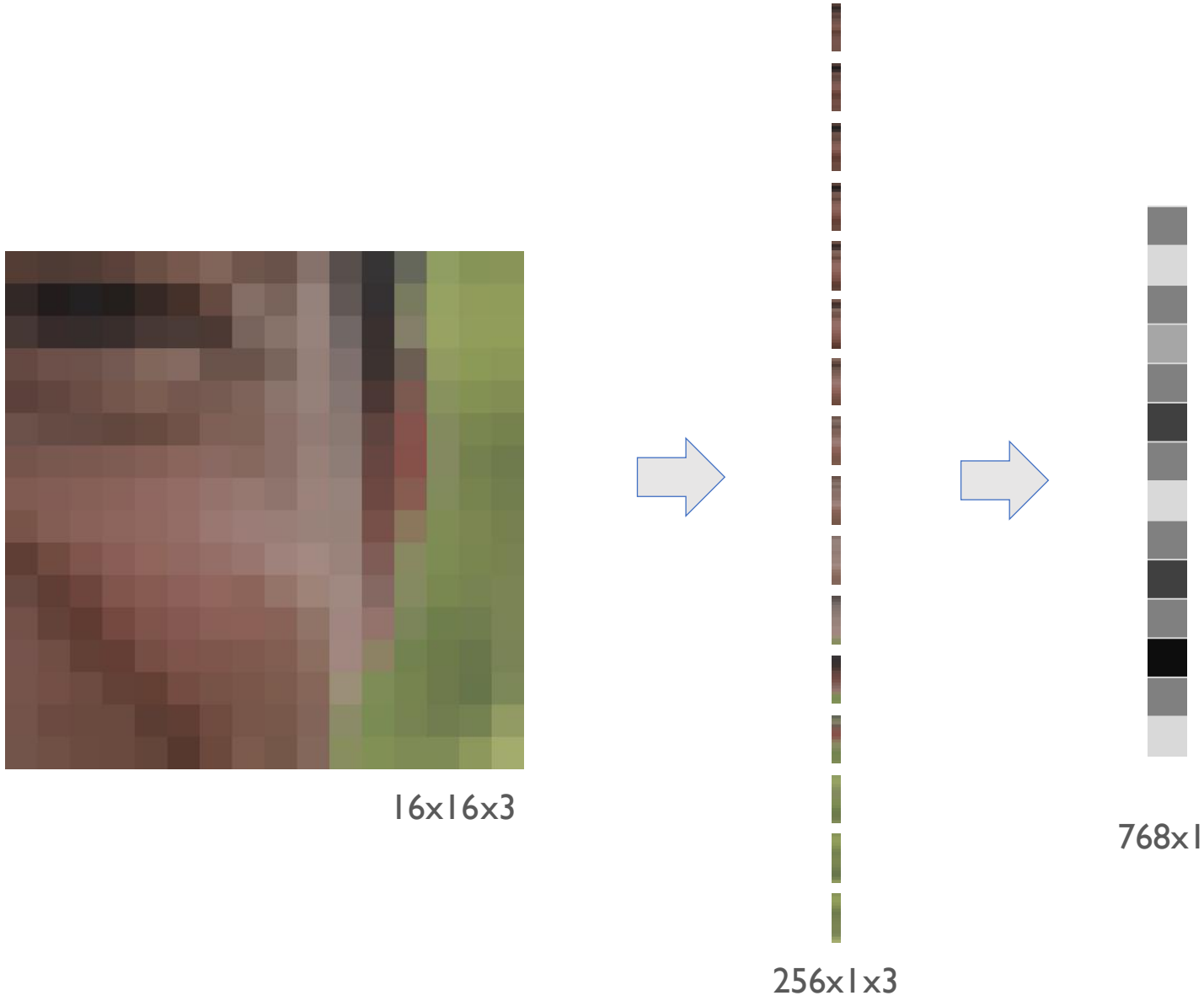
256x1x3





768x1

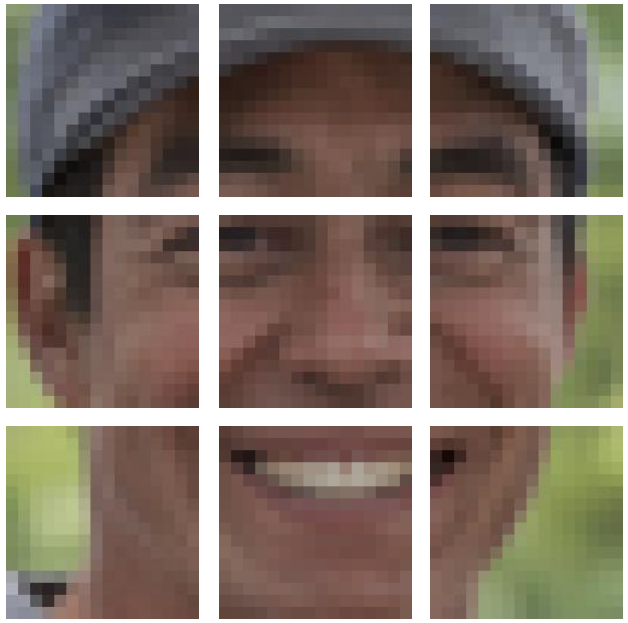
Visual Transformer (ViT)



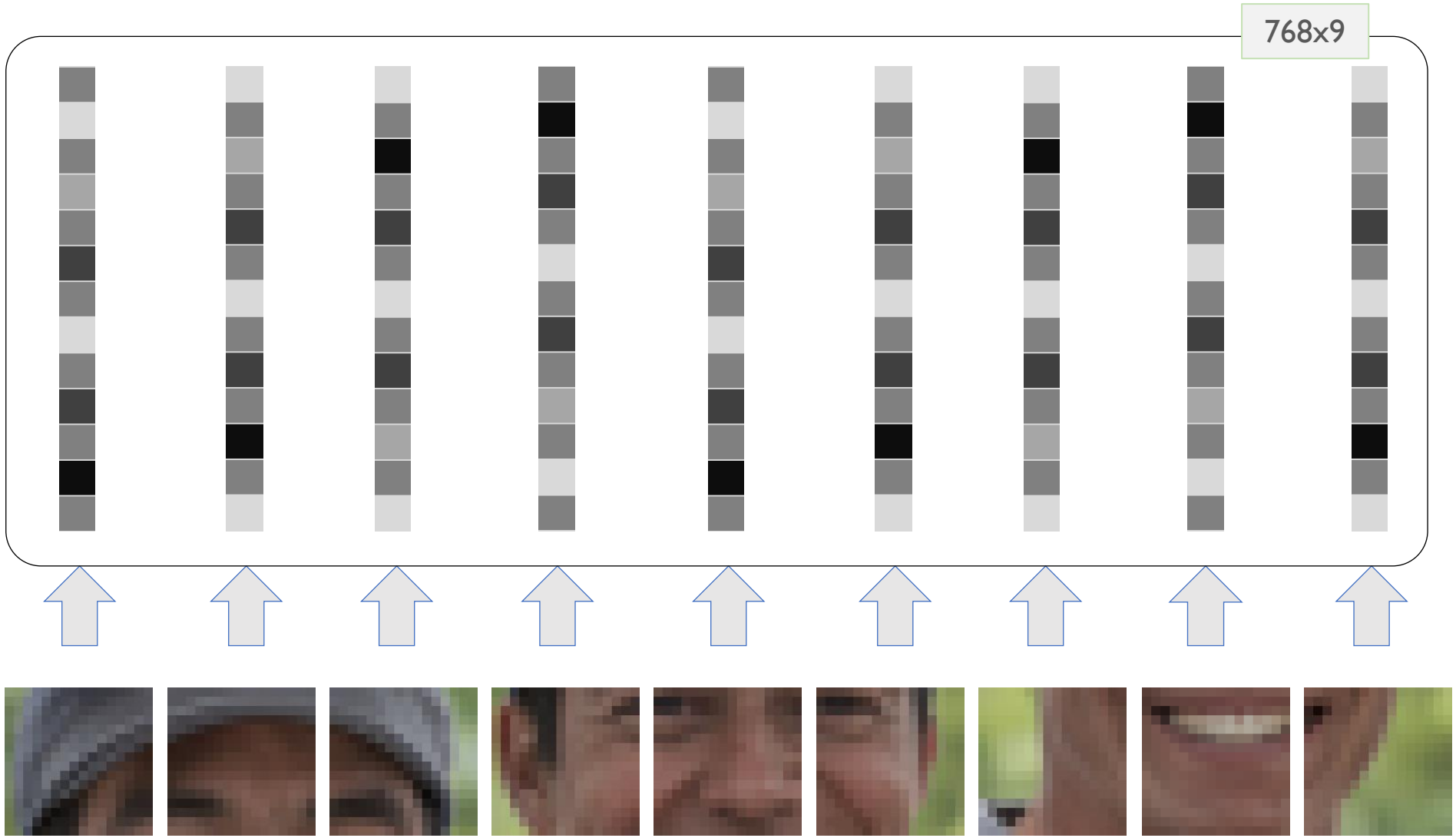
Visual Transformer (ViT)

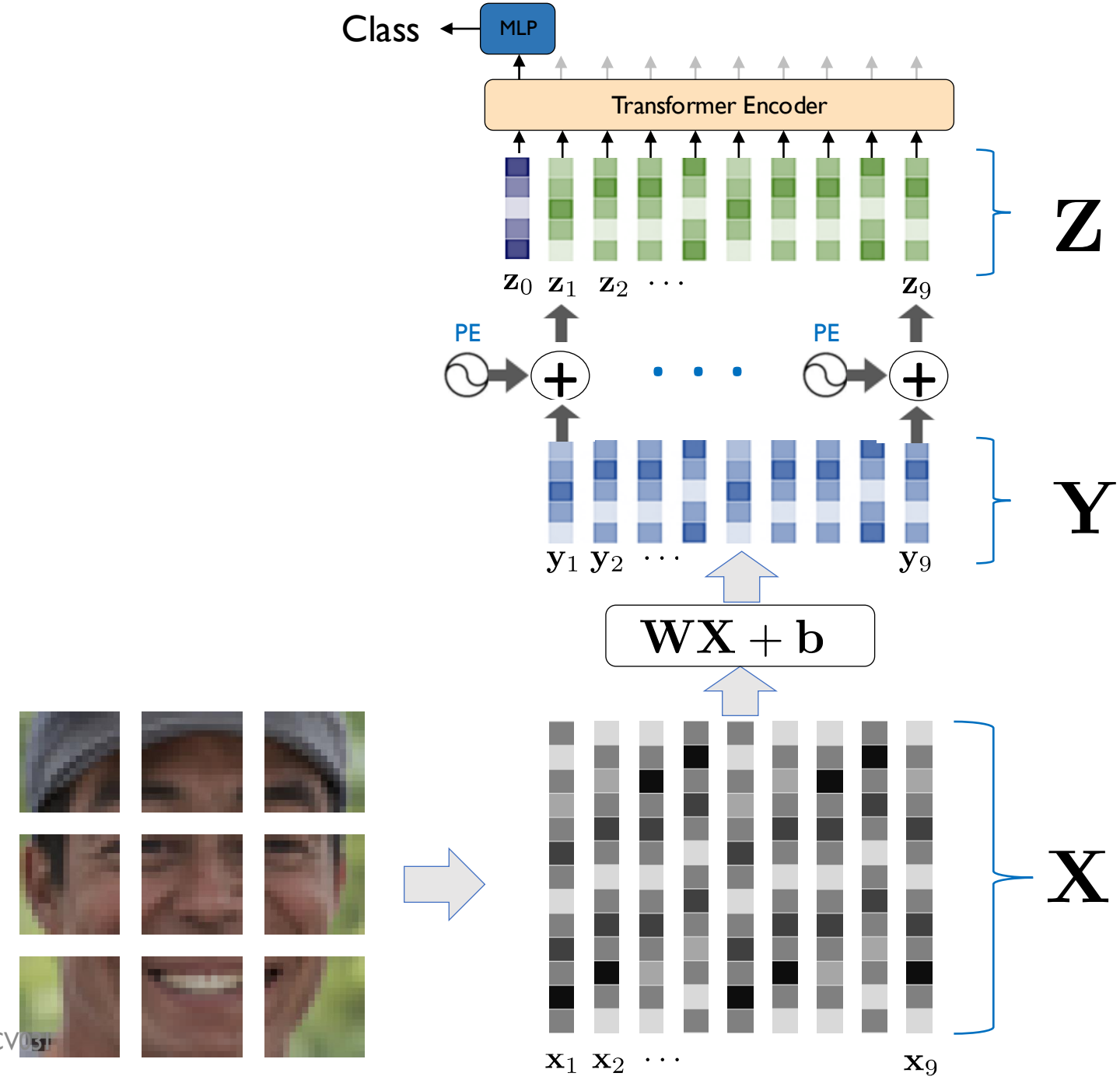


Visual Transformer (ViT)

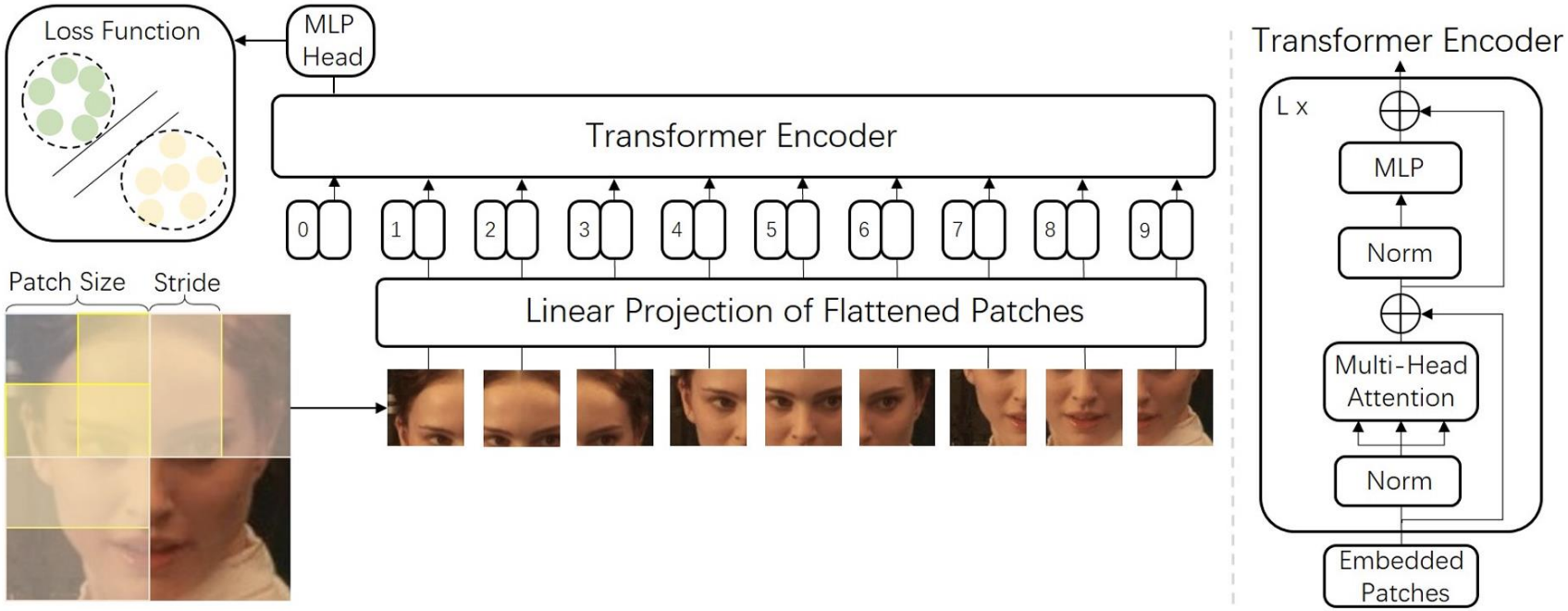


Visual Transformer (ViT)




Class
MLP
Transformer Encoder
Z
z_0 z_1 z_2 ... z_9
PE
PE
Y
y_1 y_2 ... y_9
WX + b
X
x_1 x_2 ... x_9

Visual Transformer (ViT)



Zhong, Y., & Deng, W. (2021). Face transformer for recognition. arXiv preprint arXiv:2103.14803.

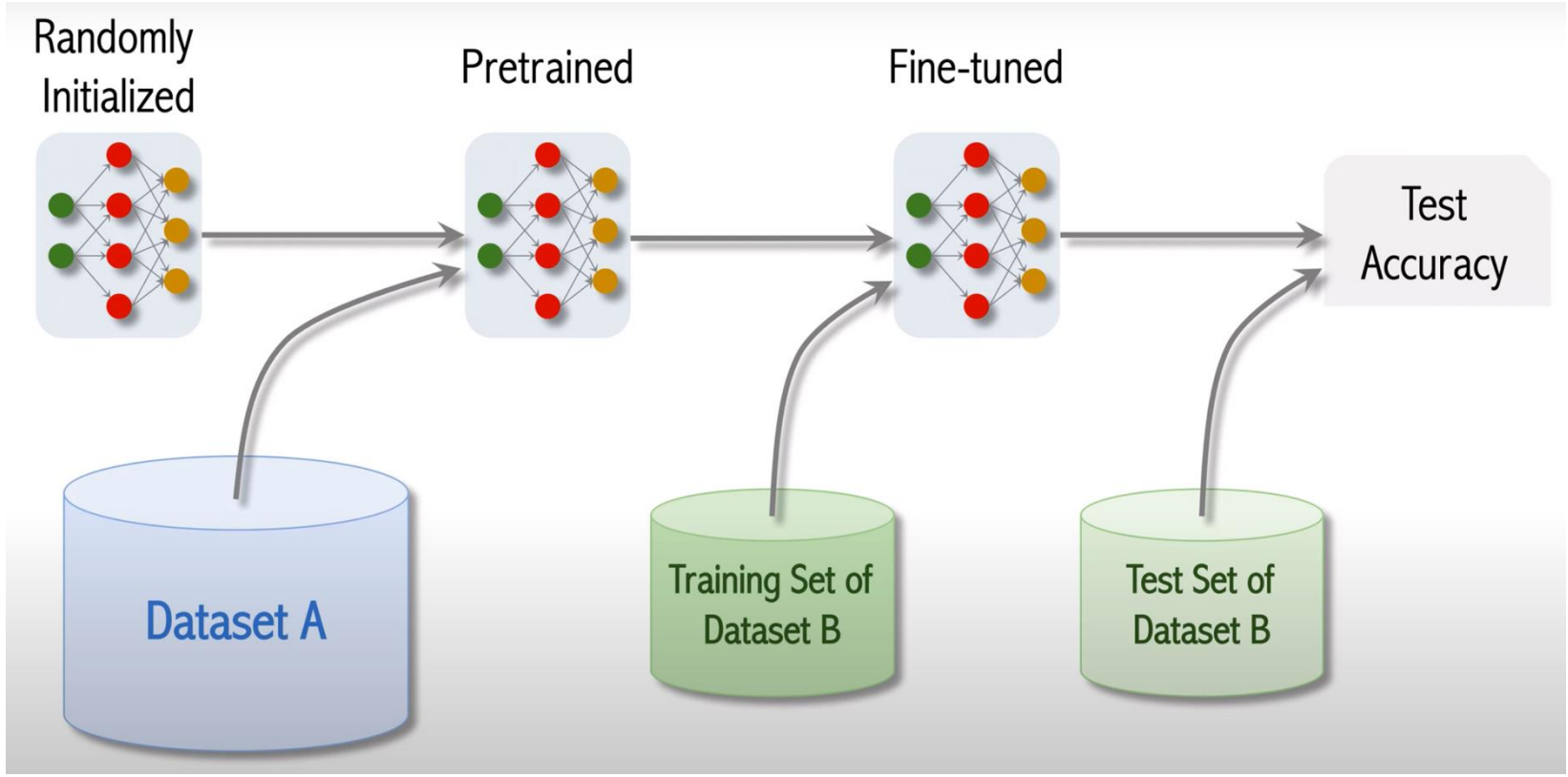
Visual Transformer (ViT)

TABLE II
PERFORMANCE ON LFW [18], SLLFW [19], CALFW [20], CPLFW [21], TALFW [22] CFP-FP [23] AND AGEDB-30 [24] DATABASES.

Training Data	Models	LFW	SLLFW	CALFW	CPLFW	TALFW	CFP-FP	AgeDB-30
CASIA-WebFace	ResNet-100 [12]	99.55	98.65	94.13	90.93	53.17	96.30	95.50
	ViT-P8S8 [1]	97.32	90.78	86.78	80.78	83.05	86.60	81.48
	ViT-P12S8	97.42	90.07	87.35	81.60	84.00	85.56	81.48
MS-Celeb-1M	ResNet-100 [12]	99.82	99.67	96.27	93.43	64.88	96.93	98.27
	ViT-P8S8 [1]	99.83	99.53	95.92	92.55	74.87	96.19	97.82
	T2T-ViT [5]	99.82	99.63	95.85	93.00	71.93	96.59	98.07
	ViT-P10S8	99.77	99.63	95.95	92.93	72.95	96.43	97.83
	ViT-P12S8	99.80	99.55	96.18	93.08	70.13	96.77	98.05

Zhong, Y., & Deng, W. (2021). Face transformer for recognition. arXiv preprint arXiv:2103.14803.

Visual Transformer (ViT)



Shusen Wang: Vision Transformer for Image Classification. https://youtu.be/HZ4j_U3FC94?si=pII9zvAlxx2eIYgq

Visual Transformer (ViT)

	# of Images	# of Classes
ImageNet (Small)	1.3 Million	1 Thousand
ImageNet-21K (Medium)	14 Million	21 Thousand
JFT (Big)	300 Million	18 Thousand

Shusen Wang: Vision Transformer for Image Classification. https://youtu.be/HZ4j_U3FC94?si=pII9zvAlxx2eIYgq

Visual Transformer (ViT): Example



On Skin Lesion Recognition Using Deep Learning: 50 Ways to Choose Your Model

Domingo Mery[1(✉)], Pamela Romero[1], Gabriel Garib[1], Alma Pedro[1], Maria Paz Salinas[2], Javiera Sepulveda[2], Leonel Hidalgo[2], Claudia Prieto[3], and Cristian Navarrete-Dechent[2]

[1] Departments of Computer Science, Pontificia Universidad Catolica de Chile, Santiago, Chile
domingo.mery@uc.cl
[2] Departments of Dermatology, Pontificia Universidad Catolica de Chile, Santiago, Chile
[3] Departments of Electrical Engineering, Pontificia Universidad Catolica de Chile, Santiago, Chile

Abstract. Skin cancer is a highly relevant health problem around the world. The World Health Organization (WHO) reports that one-third of the diagnosed cancers are skin cancer. It is well known that early detection of skin cancer significantly increases the prognosis of patients. In many cases, however, the absence of clinical devices and qualified experts makes this task very difficult. To overcome this problem, in the last years advanced deep learning techniques have been proposed to recognize skin cancer automatically showing promising results. In this work, we evaluate 50 deep learning approaches on the well-known HAM10000 dataset of dermatoscopic images from seven different skin lesion categories. The approaches have been evaluated using the same experimental protocol. In our

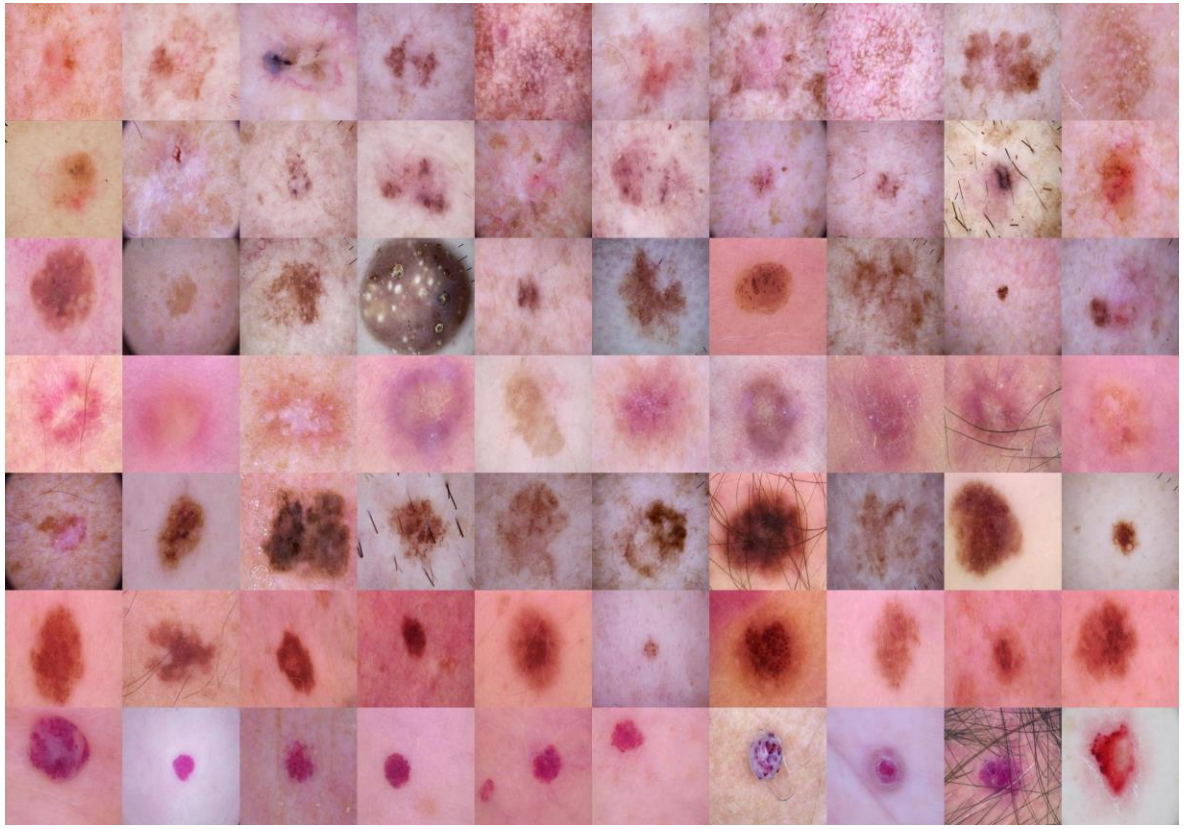
Mery, Domingo, Pamela Romero, Gabriel Garib, Alma Pedro, Maria Paz Salinas, Javiera Sepulveda, Leonel Hidalgo, Claudia Prieto, and Cristian Navarrete-Dechent. "On Skin Lesion Recognition Using Deep Learning: 50 Ways to Choose Your Model." In Pacific-Rim Symposium on Image and Video Technology, pp. 103-116. Cham: Springer International Publishing, 2022.

Evaluated Methods

50 models from
15 different architectures

Database

Class	Acronym	Name
0	akiec	Actinic Keratosis
1	bcc	Basal Cell Carcinoma
2	bkl	Benign Keratosis
3	df	Dermatofibroma
4	mel	Melanoma
5	nv	Melanocytic Nevus
6	vasc	Vascular Lesion



ISIC Dataset – HAM 10000

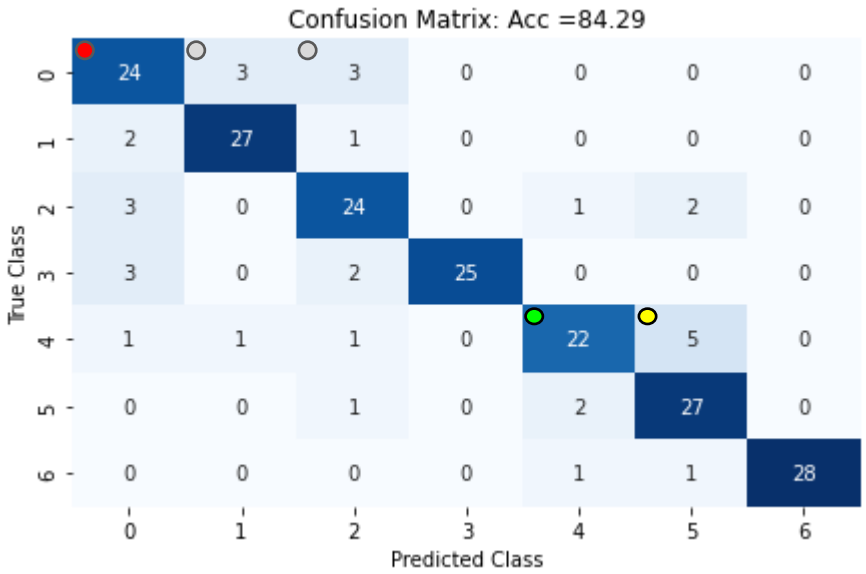
Number of images for training, validation and testing of each class

Subset	class → 0 akiec	1 bcc	2 bkl	3 df	4 mel	5 nv	6 vasc	Total
Training:	210	262	525	70	525	1050	95	2737
Validation:	30	38	75	10	75	150	14	392
Testing:	30	30	30	30	30	30	30	210
Total:	270	330	630	110	630	1230	139	3339

Results
for the
50 models

Year	#	Method
2010	1	ConvNet 5 × 80 [30]
	2	ConvNet 5 × 128 [30]
	3	ConvNet 6S × 224 [30]
	4	ConvNet 6M × 224 [30]
	5	ConvNet 6L × 224 [30]
2014	6	AlexNet [29]
	7	GoogLeNet [47]
	8	VGG11-bn [46]
	9	VGG13-bn [46]
	10	VGG16-bn [46]
	11	VGG19-bn [46]
	12	VGG11 [46]
	13	VGG13 [46]
	14	VGG16 [46]
	15	VGG19 [46]
2015	16	ResNet-18 [22]
	17	ResNet-34 [22]
	18	ResNet-50 [22]
	19	ResNet-101 [22]
	20	ResNet-152 [22]
2016	21	ResNeXt-50 [52]
	22	DenseNet201 [25]
	23	SqueezeNet1-0 [26]
	24	Wide-ResNet-50-2 [54]
2018	25	ShuffleNet-v2-0.5 [34]
	26	ShuffleNet-v2-1.0 [34]
	27	MobileNetv2 [43]
	28	MobileNetv3-S [24]

Confusion Matrix for Best Model (ViT)



Accuracy on testing subset for each architecture

